

Digital-Twin-Verified Self-Healing of the O-RAN Open-Fronthaul Synchronization Plane

Discriminating Timing Faults from Attacks and Autonomously Recovering Synchronization Before Base-Station Failure

An AI-Native Self-Healing O-RAN Network using a Digital Twin

Author: Tanmay | Project Proposal / Synopsis | Category: Integration + Experimental Validation + Released Dataset

Abstract

In the disaggregated Open Radio Access Network (O-RAN), the radio unit (O-RU) and distributed unit (O-DU) are connected by an Ethernet-based Open Fronthaul that carries a safety-critical timing (synchronization, or S-plane) signal using IEEE-1588 Precision Time Protocol (PTP) and Synchronous Ethernet (SyncE). This timing must hold to within about one hundred nanoseconds; if it drifts, the base station degrades and can fail. Recent peer-reviewed work (TIMESAFE, ACM Transactions on Privacy and Security, 2025) demonstrated that a timing-spoofing attack can crash a production 5G O-RAN base station within roughly two seconds, and it built a machine-learning monitor that *detects* such attacks with over 97.5% accuracy. However, that work stops at detection: it does not decide *what recovery action to take*, and it does not distinguish a *malicious attack* from a *benign timing fault* (loss of GNSS reference and clock holdover, packet-delay-variation, or SyncE quality degradation) that requires a different response. This project proposes an **AI-native self-healing loop for the O-RAN S-plane** that closes this gap. Passive sync telemetry is continuously monitored; an anomaly triggers a **fault-versus-attack discriminator**; candidate recovery actions (synchronization-source failover across LLS-C1 to LLS-C4 configurations, clock holdover, isolation of a rogue master, or path rerouting) are then **verified inside a digital twin of the S-plane** that predicts which action keeps timing error within budget; and the best action is committed before the two-second failure window closes. The primary deliverables are the governed self-healing loop, a released labelled dataset of timing faults and attacks, and an experimental benchmark measuring recovery success, time-to-recover, and wrong-action rate against a detection-only baseline. The contribution is honestly positioned as **integration, experimental validation, and a released dataset** — not a new fundamental algorithm.

1. Introduction and Background

1.1 O-RAN disaggregation and the Open Fronthaul

O-RAN breaks the traditional monolithic base station into interoperable, multi-vendor components — the radio unit (O-RU), distributed unit (O-DU), and centralized unit (O-CU) — connected by open, standardized interfaces. The link between O-DU and O-RU is the **Open Fronthaul** using functional split 7.2x, carried over packet-based Ethernet / eCPRI. Unlike the older synchronous CPRI, packet transport introduces packet loss and packet-delay-variation, so tight **time synchronization** must be provided separately.

1.2 The synchronization (S-) plane and why it is fragile

O-RAN Working Group 4 defines four lower-layer-split synchronization configurations, LLS-C1 through LLS-C4, that use PTP and SyncE (and optionally a local GNSS reference at the O-RU) to keep the O-DU and O-RU clocks aligned. The time-error budget is extremely tight (on the order of one hundred nanoseconds for the tightest cases; ITU-T G.8273.2 class limits). PTP was designed for accuracy, not

security, so the S-plane is exposed to timing spoofing, replay, and denial-of-service, as well as to non-malicious faults such as loss of the GNSS reference (forcing clock holdover), congestion-induced packet-delay-variation, and SyncE quality degradation.

1.3 Self-healing and the digital twin

A **self-healing** network automatically detects, diagnoses, and recovers from faults. A **digital twin** is a live software replica of the real network, kept in sync by telemetry, that can answer *what-if* questions — for example, “if I switch the timing source now, will the clock stay within budget?” — *without* risking the live network. Pairing the two lets the network verify a recovery action in simulation before applying it. This proposal makes the S-plane the target of exactly such a twin-verified self-healing loop.

2. Problem Statement

On the O-RAN Open-Fronthaul S-plane, a synchronization anomaly may arise from two physically distinct causes that demand *opposite* responses, yet appear similar as “the clock is drifting”:

- **H0 — benign timing fault:** loss of GNSS reference and holdover, packet-delay-variation, or SyncE degradation. Correct response: ride it out (holdover) or switch to a healthy timing source.
- **H1 — malicious timing attack:** PTP spoofing (a fake grandmaster / forged Announce or Sync messages) or a replay attack. Correct response: isolate the rogue source and fail over to a trusted reference.

Problem. Given a stream of passive S-plane synchronization telemetry, (i) detect a timing anomaly, (ii) decide H0 (fault) versus H1 (attack), and (iii) select and apply a recovery action that keeps the time error within the O-RAN budget, **committing the action before the base station fails (~2 s)** — while minimizing the wrong-action rate and the mean time to recover (MTTR). Existing work detects the anomaly but leaves (ii) and (iii) undefined.

3. Motivation and Significance

- **High stakes, short fuse.** A timing attack can catastrophically fail a production O-RAN base station in about two seconds, requiring manual recovery [2]. Detection alone does not prevent the outage; an automated response does.
- **No standard yet.** The O-RAN Alliance states that no comprehensive fronthaul security standard yet exists, and the WG4 annex specifies parameters to monitor but *not* the actions to take on detection [1,2].
- **Fault and attack are conflated.** Treating every timing anomaly as an attack (or every anomaly as a fault) produces wrong recoveries; the two must be separated to act correctly.
- **Title-faithful.** The problem exercises all four pillars of the project title: it is O-RAN, it is AI-native (learned discrimination), it is self-healing (autonomous recovery), and the digital twin is load-bearing (it vets the action).

4. Research Gap (Honest Positioning)

The nearest peer-reviewed work, **TIMESAFE** [2], (a) is *detection-only* and explicitly notes that the O-RAN specification does not define the response action; (b) discriminates benign *traffic* from attack *traffic*, but does *not* separate a malicious attack from a benign timing *fault*; and (c) uses its digital twin to generate data and stage attacks, *not* as a counterfactual verifier of a recovery action. General twin-verified control in O-RAN (OpenTwin [3]) targets energy optimization and assumes fresh telemetry; conflict-mitigation work [4,5,6] gates actions on xApp-versus-xApp conflict, not on timing trust. The specific combination proposed here — **fault-versus-attack discrimination plus digital-twin-verified S-plane recovery, evaluated**

experimentally against a detection-only baseline — is, to the best of a thorough IEEE / Elsevier / MDPI / arXiv search, unoccupied. We state plainly that the *mechanisms* reused (ML detection, twin what-if) are known; the *contribution is the response step, the fault/attack split, and the experimental evidence*.

5. Related Work

5.1 Fronthaul timing security

TIMESAFE [2] demonstrates PTP spoofing/replay crashing a 5G base station in ~2 s and an ML monitor (transformer/CNN) at >97.5% detection; it releases attack scripts, data-collection tools, trained detectors, a testbed/twin, and a packet-trace dataset. Multi-layer defence and KPI-poisoning studies [13,14] characterize O-RAN control-loop poisoning threats. These define the threat and the detector; none defines the recovery.

5.2 Digital twins for O-RAN

OpenTwin [3] performs twin-driven closed-loop KPM control with fidelity-aware resynchronization, applied to energy saving. Colosseum [11] is a large-scale O-RAN digital twin with software-defined-radio in the loop. The O-RAN nGRG digital-twin research report [16] lists twin use cases and flags *missing/delayed telemetry* as a key open fidelity problem. IEEE Communications Magazine [18] frames the twin for intelligent, resilient O-RAN. Sim-to-real gap work [12] and uncertainty-aware twins [17] address twin trust in general terms.

5.3 Self-healing, anomaly detection, and action governance

Zero-shot self-healing [7], the failure-management survey [8], Hy-LIFT diagnosis [9], and Simba root-cause analysis [10] establish RAN self-healing methods and repeatedly note the scarcity of real, labelled fault data. Conflict-mitigation frameworks [4,5,6] and ARGOS anomaly guarding [15] govern or detect at the RIC/E2 layer. All operate above the fronthaul; none targets S-plane timing recovery.

5.4 Positioning summary

Work	What it does	What it does NOT do (our gap)
TIMESAFE [2]	Detects PTP attacks (>97.5%); releases data/tools/twin	No recovery action; no fault-vs-attack split; twin not used to verify a fix
OpenTwin [3]	Twin-verified control loop (energy); resync on drift	Not timing/self-healing; assumes fresh telemetry
Conflict mgmt [4,5,6]	Blocks/reconciles conflicting xApp actions	Trigger is xApp conflict, not timing trust; RIC layer, not S-plane
This project	Fault-vs-attack split + twin-verified S-plane recovery + dataset	— (the proposed contribution)

6. Research Objectives

- **O1.** Build a monitored O-RAN S-plane (emulated fronthaul with PTP/SyncE) and reproduce timing faults and attacks with timestamped, auto-labelled logging.
- **O2.** Design and train a **fault-versus-attack discriminator** on sync telemetry (offset, path delay, PDV, PTP message regularity, SyncE quality).
- **O3.** Build a **digital twin of the S-plane** that forecasts time-error drift under a candidate recovery action versus the safe default.
- **O4.** Implement a **governed recovery controller** that commits the twin-preferred action within the failure window (source failover / holdover / isolate / reroute).
- **O5.** Release a **labelled fault/attack dataset + benchmark** and evaluate the full loop against a detection-only baseline.

7. Proposed Methodology

The self-healing loop runs continuously over passive S-plane telemetry and proceeds in four governed stages:

- **Stage 1 — Sync-anomaly detection.** A lightweight detector flags windows whose timing behaviour departs from the learned healthy baseline (offset excursion, path-delay spike, PDV burst).
- **Stage 2 — Fault-vs-attack discrimination.** The flagged window is classified H0 (fault) or H1 (attack). Rationale: attacks break *PTP protocol regularities* (duplicated/anomalous sequence IDs, forged Announce, implausibly abrupt offset steps), whereas faults degrade *gradually and physically* (holdover ramp, congestion-correlated PDV). This is a two-hypothesis test, not a single anomaly bit.
- **Stage 3 — Digital-twin counterfactual verification.** For each admissible recovery action, the S-plane twin rolls the clock dynamics forward and predicts the resulting time error; an action is admissible for commit only if the twin predicts it keeps time error within budget and beats the safe default (holdover).
- **Stage 4 — Governed recovery commit.** The controller applies the best verified action and logs an auditable reason. A hard real-time constraint bounds Stages 1-4 to complete well inside the ~2 s failure window; if the twin is uncertain, the loop defaults to the safest action.

Design principle. The digital twin is *load-bearing at decision time*: it does not merely generate data, it decides whether a recovery is safe to apply. This is precisely the step the detection-only state of the art leaves undefined.

8. System Design and Architecture

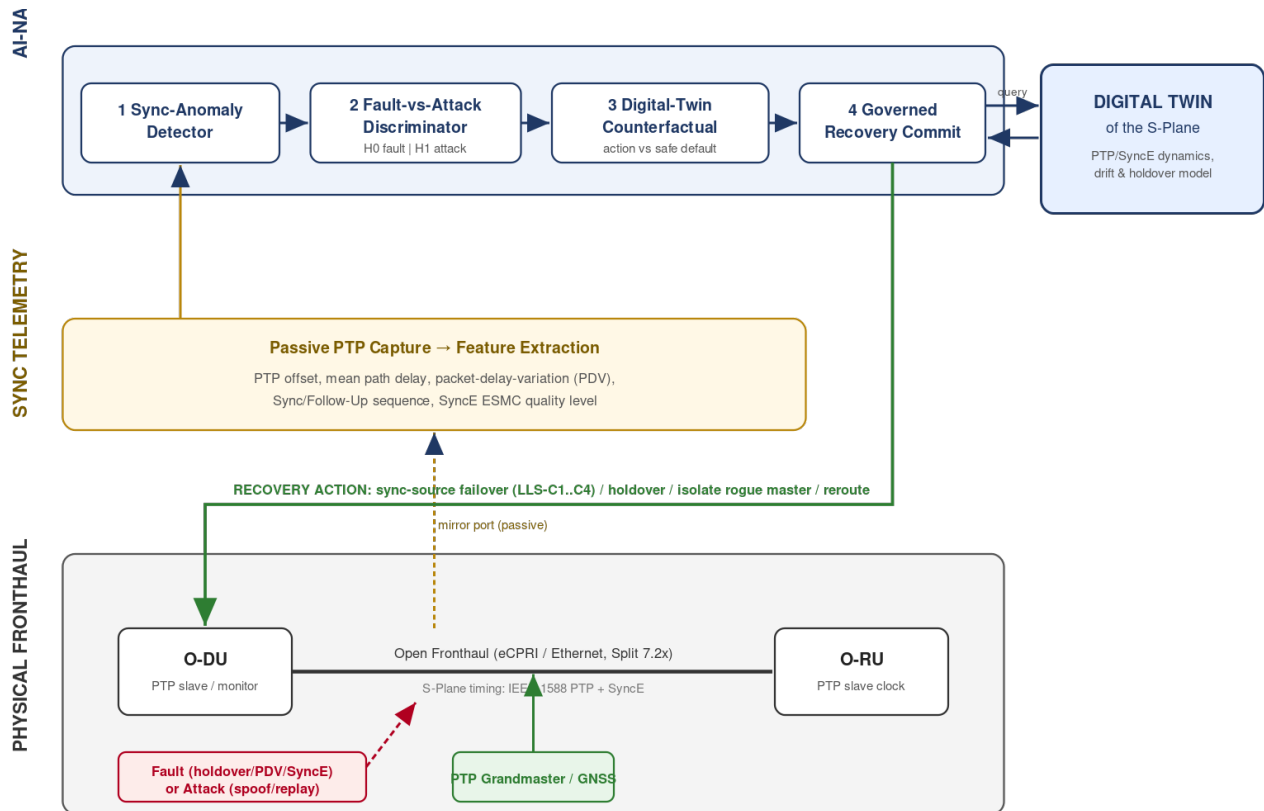


Figure 1. Architecture of the digital-twin-verified S-plane self-healing loop.

Physical fronthaul (bottom). O-DU and O-RU exchange PTP/SyncE timing over the Split-7.2x Open Fronthaul; a PTP grandmaster (optionally GNSS-disciplined) is the reference. A fault source or attacker

perturbs the timing. **Sync telemetry (middle)**. A passive mirror-port capture at the O-DU extracts features (offset, mean path delay, PDV, PTP sequence/message type, SyncE ESMC quality) without interfering with timing. **Self-healing loop (top)**. Detector → discriminator → twin verifier → governed commit. **Digital twin (right)**. Queried by the verifier to forecast drift for each candidate action. **Recovery (green path)**. The committed action reconfigures the O-DU/O-RU timing (failover across LLS-C1..C4, holdover, isolate rogue master, or reroute).

9. Digital-Twin Design

The S-plane twin is a compact dynamical model (not a full-PHY simulator) of clock synchronization: it models the servo that disciplines the slave clock to PTP offset and path-delay inputs, the holdover behaviour when the reference is lost, and the time-error accumulation under a given timing source. It is calibrated on healthy telemetry and reports a **fidelity score** so the controller can discount its advice when it is out of sync — an explicit nod to the O-RAN nGRG open problem of missing/delayed twin telemetry [16]. Given a candidate action, the twin returns a predicted time-error trajectory used for the commit decision.

10. Fault and Attack Model

Class	Instance	Signature the loop exploits	Correct recovery
H0 fault	GNSS loss / holdover	Gradual offset ramp; reference-status flag	Ride holdover or failover to healthy source
H0 fault	Packet-delay-variation	PDV correlated with load; symmetric	Reroute / smooth; hold if transient
H0 fault	SyncE degradation	ESMC quality-level drop	Failover to alternate sync source
H1 attack	PTP spoofing	Forged Announce/Sync; abrupt offset step	Isolate rogue master; failover to trusted ref
H1 attack	Replay	Duplicate / out-of-order sequence IDs	Isolate path; failover; alarm

11. Recovery Action Space

- **Sync-source failover** across LLS-C1..C4 (e.g., from network PTP to a local GNSS reference at the O-RU, LLS-C4).
- **Holdover** — rely on the disciplined internal oscillator for a bounded interval while the anomaly is assessed.
- **Isolate rogue master** — reject the spoofing source (e.g., BMCA/allow-list enforcement).
- **Reroute** — select an alternate fronthaul path with lower delay variation.
- **Safe default** — the conservative fallback the twin-verified action must beat.

12. Build Workflow / Implementation Plan

Phase	Goal	Key tools / assets	Exit criterion
P1 Monitored S-plane	Emulated fronthaul with PTP+SyncE and passive capture	linuxptp (ptp4l), SyncE, tc netem, Wireshark; TIMESAFE s-plane_security repo [2]	Live sync telemetry stream captured and parsed
P2 Fault/attack injection	Timestamped, auto-labelled faults and attacks	tc netem (PDV/holdover), spoof/replay scripts [2]	Labelled dataset of H0 and H1 events
P3 Discriminator	Train + evaluate fault-vs-attack classifier	Python (scikit-learn/torch); features from P1	ROC / confusion beating detection-only baseline
P4 S-plane twin	Drift forecast per candidate action	Clock-servo + holdover model; calibrate on healthy data	Twin predicts time error within tolerance
P5 Governed recovery	Commit twin-verified action in-window	Controller + action API to ptp4l config	Recovery applied < failure window in tests

Phase	Goal	Key tools / assets	Exit criterion
P6 Benchmark + release	Metrics vs baselines; publish dataset	Evaluation harness; dataset + code	Reproducible results table + released artifacts

13. Validation Ladder and Datasets

- **(A) Released traces.** Start from the TIMESAFE packet-trace dataset and trained detectors [2] as baseline and sanity check.
- **(B) Emulation.** linuxptp + SyncE over an Ethernet link with tc netem for PDV/holdover and the released spoof/replay scripts — no special hardware needed.
- **(C) Optional hardware.** PTP-capable NIC / boundary-clock switch and an O-RU for full-realism time-error measurement (stretch tier).
- **Honest caveat.** Tiers A-B are fully feasible on a standard Linux machine; nanosecond-accurate realism (C) needs timing hardware and is optional.

14. Evaluation Metrics

Metric	Meaning	Goal
Discrimination accuracy / ROC	Correctly separating H0 fault from H1 attack	High; beat single-bit detector
Recovery success rate	Timing restored within budget after action	Maximize
Time error under budget	Peak/steady time error vs G.8273.2 limit	Kept within limit
MTTR vs failure window	Time to recover relative to the ~2 s crash	Recover before failure
Wrong-action rate	Harmful/incorrect recovery chosen	Minimize

Baselines: (i) TIMESAFE-style detection-only with no response; (ii) naive always-holdover; (iii) always-failover. The headline result is the reduction in wrong-action rate and outage avoided by the twin-verified loop.

15. Expected Outcomes and Contributions

- A working **digital-twin-verified S-plane self-healing loop** (discriminate → verify → recover).
- A **released, labelled dataset** of S-plane timing faults and attacks with an open benchmark.
- Quantified evidence that twin-verified recovery **avoids outages** a detection-only system cannot.
- A reproducibility bundle (scripts, twin, models) extending peer-reviewed released artifacts [2].

16. Novelty Statement (Stated Honestly)

This is an **integration + experimental-validation + dataset** contribution, the class routinely published at O-RAN conferences/workshops. The reused mechanisms (ML anomaly detection; digital-twin what-if verification) are established. The genuine contribution is threefold: (1) the **recovery/response step** for S-plane timing that the peer-reviewed state of the art explicitly leaves undefined; (2) the **fault-versus-attack discrimination** that lets the response be correct; and (3) the **twin used as a decision-time verifier** of that response, with released experimental evidence. We do not claim a new fundamental algorithm.

17. Risks and Mitigations

Risk	Mitigation
No PTP timing hardware for ns-accurate realism	Validate on released traces + emulation (tiers A-B); mark hardware as optional stretch
Twin fidelity too low to trust a recovery	Fidelity score gates the twin's advice; default to safest action when uncertain
Real-time budget (~2 s) hard to meet	Keep detector/twin lightweight; measure and report per-stage latency
Reviewer: 'detection is known'	Position honestly as response + fault/attack split + evidence; cite gap in [2]
Fault and attack signatures overlap	Use PTP protocol-regularity features that faults do not exhibit

18. Indicative Timeline

Milestone	Phase(s)	Approx. duration
Monitored S-plane + capture	P1	2-3 weeks
Fault/attack dataset	P2	2-3 weeks
Discriminator trained + evaluated	P3	3-4 weeks
S-plane digital twin	P4	3-4 weeks
Governed recovery + integration	P5	3-4 weeks
Benchmark, dataset release, write-up	P6	3-4 weeks

References

- [1] O-RAN ALLIANCE WG4, "Control, User and Synchronization Plane Specification (O-RAN.WG4.CUS.0)," incl. LLS-C1–C4 synchronization configurations (Sec. 11).
- [2] J. Groen, S. Di Valerio, I. Karim, D. Villa, et al., "TIMESAFE: Timing Interruption Monitoring and Security Assessment for Fronthaul Environments," ACM Transactions on Privacy and Security, vol. 28(5), 2025. DOI 10.1145/3775060 (arXiv:2412.13049). Code: github.com/genesys-neu/s-plane_security.
- [3] "OpenTwin: Digital Twin Driven Closed Loop KPM Inference and Control for Open RAN," arXiv:2605.24662, 2026.
- [4] A. Wadud, F. Golpayegani, N. Afraz, "Conflict Management in the Near-RT-RIC of Open RAN: A Game Theoretic Approach," IEEE iThings/GreenCom/CPSCoM, 2023.
- [5] N. Varshney, C. Puligheddu, A. Badawy, C. F. Chiasserini, "XAI4C: Explainable AI for Conflict Detection and Mitigation in O-RAN Near-RT RIC," IEEE Vehicular Technology Magazine, 2025. DOI 10.1109/MVT.2025.3625693.
- [6] "Conflict Mitigation Framework and Conflict Detection in O-RAN Near-RT RIC," IEEE (doc 10121578), 2023 (arXiv:2305.07117).
- [7] H. Bilen et al., "KDN-Driven Zero-Shot Learning for Intelligent Self-Healing in 6G Small Cell Networks," Ad Hoc Networks, vol. 178, 103984, 2025 (Elsevier).
- [8] "Failure Management in 5G RAN: Challenges and Open Research Lines," survey.
- [9] A. D. Salman et al., "Hy-LIFT: Hybrid LLM-Assisted Fault Diagnosis Framework for 5G/6G Networks," MDPI Computers, 14(12):551, 2025.
- [10] "Simba: Root Cause Analysis of Anomalies in 5G RAN using GNN + Transformer," arXiv:2406.15638.
- [11] L. Bonati et al., "Colosseum: The Open RAN Digital Twin," arXiv:2404.17317, 2024.
- [12] "How to Bridge the Sim-to-Real Gap in Digital Twin-Aided Telecommunication Networks," arXiv:2507.07067, 2025.
- [13] "KPI Poisoning: An Attack in Open RAN Near Real-Time Control Loop," arXiv:2505.05537.
- [14] "Towards a Multi-Layer Defence Framework for Securing Near-Real-Time Operations in Open RAN," arXiv:2512.01596.
- [15] "ARGOS: Anomaly Recognition and Guarding through O-RAN Sensing," arXiv:2506.06916.
- [16] O-RAN nGRG, "Research Report on Digital Twin RAN Use Cases," RS-01, 2024.
- [17] "Uncertainty-Aware Digital Twins: Robust Model Predictive Control using Time-Series Deep Quantile Learning," arXiv:2501.10337, 2025.

- [18] “Digital Twin for Open RAN: Towards Intelligent and Resilient 6G Radio Access Networks,” IEEE Communications Magazine, 2023.
- [19] IEEE 1588-2019 (PTP); ITU-T G.8275.1 (PTP telecom profile), G.8273.2 (time-error limits for telecom boundary clocks).
- [20] srsRAN Project and FlexRIC (EURECOM) — open-source O-RAN gNB and Near-RT RIC.